🧼 Germ Busters

Name: Class: Date:

Activity 1 · Hands-on · 🏫 In class · ⏱ 30 minutes

✨ Glitter Germs: Why Handwashing Works

You will learn: Many infectious diseases spread by touch; washing with soap for 20 seconds removes most germs.

YOU NEED

- Glitter or turmeric powder mixed with a little oil (pretend germs)
- Soap, water, a towel
- Timer

⚠️ **Safety:** Keep glitter away from eyes. Use turmeric if anyone is sensitive to glitter.

STEPS

1. Rub "germs" on one volunteer's hands.
2. The volunteer shakes hands with 3 people, who then touch a pen, a desk and a book.
3. Count how many people and objects now have "germs".
4. Three volunteers wash differently: dry tissue wipe; water only for 5 s; soap and water for 20 s.
5. Compare how much glitter is left on each.

OBSERVATION TABLE

Cleaning method	Germs left (lots / some / almost none)

QUESTIONS

1. Which method removed the most "germs"?

2. Name two diseases that spread through contaminated food or water.

3. Why is 20 seconds of scrubbing recommended?

WHAT IS HAPPENING? — THE SCIENCE

Typhoid, amoebiasis, ascariasis, the common cold and many others spread by contaminated hands, food or surfaces. Soap breaks down oily dirt and the lipid coats of many germs; rubbing for 20 seconds lifts them off. Personal hygiene is the first step in preventing infectious diseases.

Key terms: infectious disease · pathogen · hygiene

Activity 2 · Hands-on · 🌳 Outdoors · ⏱ 1 hour

🦟 Mosquito Breeding Survey

> **You will learn:** Mosquitoes breed in still water; removing breeding sites prevents malaria, dengue and filariasis.

YOU NEED

- Notebook
- Torch
- Gloves

> ⚠️ **Safety:** Wear gloves and do not touch dirty water with bare hands. Ask an adult before emptying any container.

STEPS

1. Check around your home: coolers, flower-pot trays, old tyres, buckets, coconut shells, water tanks.
2. In each, look for wriggling larvae (wrigglers).
3. Record the place, whether water was present and whether larvae were seen.
4. With permission, empty, cover or clean each site.
5. Make a small poster: "No still water — no mosquitoes".

OBSERVATION TABLE

Place checked	Still water?	Larvae seen?	Action taken

QUESTIONS

1. Which mosquito spreads malaria? Which spreads filariasis?

2. Name a fish used to control mosquito larvae.

3. Why is emptying coolers weekly important?

WHAT IS HAPPENING? — THE SCIENCE

Female Anopheles mosquitoes spread Plasmodium (malaria); Aedes spreads dengue and chikungunya; Culex spreads Wuchereria (filariasis). All lay eggs in still water. Removing breeding sites and using Gambusia fish in ponds are effective controls.

Key terms: **vector** **Anopheles** **Plasmodium** **Gambusia**

Digital activity

💻 Who Causes It?

> Tap a disease, then tap the type of pathogen.

Do this digital activity in the BAIO app: **https://narendra-bio-sir.github.io/BAIO-Learning-System-By-Narendra-Sir/activities.html#pu2-ch7/2**

BACTERIUM · VIRUS · PROTOZOAN · HELMINTH (WORM) · FUNGUS

Bacterium	Virus	Protozoan	Helminth (worm)	Fungus

Typhoid · Pneumonia (Streptococcus) · Common cold · AIDS · Malaria · Amoebiasis · Ascariasis · Filariasis · Ringworm

> **You will learn:** Knowing the pathogen guides treatment: antibiotics act on bacteria, not on viruses.

THINK & WRITE

1. Why don't antibiotics cure a cold?

2. Which of these diseases is spread by mosquitoes?